

## Session 12 (AIML) – Assignment Question

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DATE: 01.07.2026, WEDNESDAY

SANTOSH KALE

Q1:

TASK 14.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/TASK 14.py (3.13.5)

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**# Creating a simple 1D array**

```
arr1 = np.array([10, 20, 30, 40, 50])
```

```
print("1D Array:")
```

```
print(arr1)
```

**# Printing shape and datatype**

```
print("Shape:", arr1.shape)
```

```
print("Data Type:", arr1.dtype)
```

**# \_\_\_\_\_**

**# Part (b) - Create 2D Array**

**# \_\_\_\_\_**

**# Creating a 3x3 matrix**

```
arr2 = np.array([
[1, 2, 3],
[4, 5, 6],
[7, 8, 9]
])
```

```
print("\n2D Array:")
```

```
print(arr2)
```

**# Printing shape and datatype**

```
print("Shape:", arr2.shape)
```

```
print("Data Type:", arr2.dtype)
```

**# End of Program**

```
*IDLE Shell 3.13.5*
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Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/TASK 14.py ===
1D Array:
[10 20 30 40 50]
Shape: (5,)
Data Type: int64

2D Array:
[[1 2 3]
 [4 5 6]
 [7 8 9]]
Shape: (3, 3)
Data Type: int64
>>> |
```

Q2:

TASK 14.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/TASK 14.py (3.13.5)

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# Q2. (np.zeros() and np.ones())

import numpy as np

# \_\_\_\_\_

# Part (a) - 1D array of zeros

# \_\_\_\_\_

zero1 = np.zeros(8)

print("\n1D Array of Zeros:")

print(zero1)

# \_\_\_\_\_

# Part (b) - 4x4 array of ones

# \_\_\_\_\_

one\_array = np.ones((4, 4))

print("\n4x4 Array of Ones:")

print(one\_array)

# \_\_\_\_\_

# Part (c) - 3x3 matrix of zeros

# \_\_\_\_\_

zero\_matrix = np.zeros((3, 3))

print("\n3x3 Matrix of Zeros:")

print(zero\_matrix)

```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win:
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/TASK 14.py ===
1D Array of Zeros:
[0. 0. 0. 0. 0. 0. 0. 0.]

4x4 Array of Ones:
[[1. 1. 1. 1.]
 [1. 1. 1. 1.]
 [1. 1. 1. 1.]
 [1. 1. 1. 1.]]

3x3 Matrix of Zeros:
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
>>>
```

Q3:

```
TASK 14.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/TASK 14.py (3.13.5)
File Edit Format Run Options Window Help

# Q3. (np.arange())

import numpy as np

# a) Numbers from 0 to 20

arr1 = np.arange(0, 21, 1)

print("0 to 20:")
print(arr1)

# b) Even numbers from 10 to 50

arr2 = np.arange(10, 51, 2)

print("\nEven Numbers:")
print(arr2)

# c) Numbers from 5 to 100 with step 5

arr3 = np.arange(5, 101, 5)

print("\nStep of 5:")
print(arr3)

# End
|

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Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win
Enter "help" below or click "Help" above for more information.

>>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/TASK 14.py ===
0 to 20:
[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20]

Even Numbers:
[10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50]

Step of 5:
[ 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90
 95 100]
>>> |
```

Q4:

```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
Enter "help" below or click "Help" above for more information.

>>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/TASK 14.
Array 1:
[0.    0.55555556 1.11111111 1.66666667 2.22222222 2.77777778
 3.33333333 3.88888889 4.44444444 5.    ]
Length: 10

Array 2:
[-10.    -8.57142857 -7.14285714 -5.71428571 -4.28571429
 -2.85714286 -1.42857143  0.    1.42857143  2.85714286
  4.28571429  5.71428571  7.14285714  8.57142857 10.    ]
Length: 15

>>> |
```

# Q4. (np.linspace())

import numpy as np

# a) 10 equally spaced numbers between 0 and 5

arr1 = np.linspace(0, 5, 10)

print("Array 1:")

print(arr1)

print("Length:", len(arr1))

# b) 15 equally spaced numbers between -10 and 10

arr2 = np.linspace(-10, 10, 15)

print("\nArray 2:")

print(arr2)

print("Length:", len(arr2))

# End

|

Q5:

TASK 14.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/TASK 14.py (3.1

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# Q5. (Random Arrays)

import numpy as np

# a) 10 random numbers between 0 and 1

arr1 = np.random.rand(10)

print("Random Numbers:")

print(arr1)

# b) 3x3 matrix from normal distribution

arr2 = np.random.randn(3,3)

print("\nRandom Normal Matrix:")

print(arr2)

# c) Random integers (4x5) between 10 and 50

arr3 = np.random.randint(10, 51, (4,5))

print("\nRandom Integer Matrix:")

print(arr3)

# End

# Q5. (Random Arrays)

import numpy as np

# a) 10 random numbers between 0 and 1

arr1 = np.random.rand(10)

print("Random Numbers:")

print(arr1)

print("\nRandom Normal Matrix:")

print(arr2)

# c) Random integers (4x5) between 10 and 50

arr3 = np.random.randint(10, 51, (4,5))

print("\nRandom Integer Matrix:")

print(arr3)

# End

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
Enter "help" below or click "Help" above for more information.

>>>

=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/TASK 14.py ===

Random Numbers:
[0.67723624 0.62289442 0.60144318 0.84112502 0.86956288 0.43797648
 0.54787484 0.21925772 0.14667955 0.61117472]

Random Normal Matrix:
[[-0.64634495 -1.10834137  1.33372105]
 [ 0.22835697 -0.85099778 -0.13189605]
 [-0.30369029  1.62784304  0.07565337]]

Random Integer Matrix:
[[45 43 15 28 23]
 [23 13 12 44 10]
 [37 25 40 20 25]
 [17 13 13 19 12]]

Random Numbers:
[0.45487082 0.60365153 0.33640343 0.14565205 0.39690742 0.07124739
 0.99286679 0.38918706 0.00827671 0.15975984]

Random Normal Matrix:
[[-1.27689865  0.39849916  0.01006 ]
 [-1.90551809  0.40793912 -1.59001451]
 [-0.64356576  0.38054954  0.98295919]]

Random Integer Matrix:
[[44 24 47 37 32]
 [37 18 12 45 28]
 [45 44 17 49 47]
 [10 14 46 38 34]]

>>>
```

Q6:

TASK 14.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/TASK 14.py (3.13.5)

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**# Q6. (Vectors and Operations)**

**import numpy as np**

**# Creating vectors**

**v1 = np.array([2, 4, 6, 8])**

**v2 = np.array([1, 3, 5, 7])**

**print("Vector 1:", v1)**

**print("Vector 2:", v2)**

**# Addition**

**print("\nAddition:")**

**print(v1 + v2)**

**# Subtraction**

**print("\nSubtraction:")**

**print(v1 - v2)**

**# Multiplication**

**print("\nElement-wise Multiplication:")**

**print(v1 \* v2)**

**# Dot Product**

**print("\nDot Product:")**

**print(np.dot(v1, v2))**

**# End**

Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]  
Enter "help" below or click "Help" above for more information.

&gt;&gt;&gt;

=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/TASK 14.py ===

Vector 1: [2 4 6 8]

Vector 2: [1 3 5 7]

Addition:

[ 3 7 11 15]

Subtraction:

[1 1 1 1]

Element-wise Multiplication:

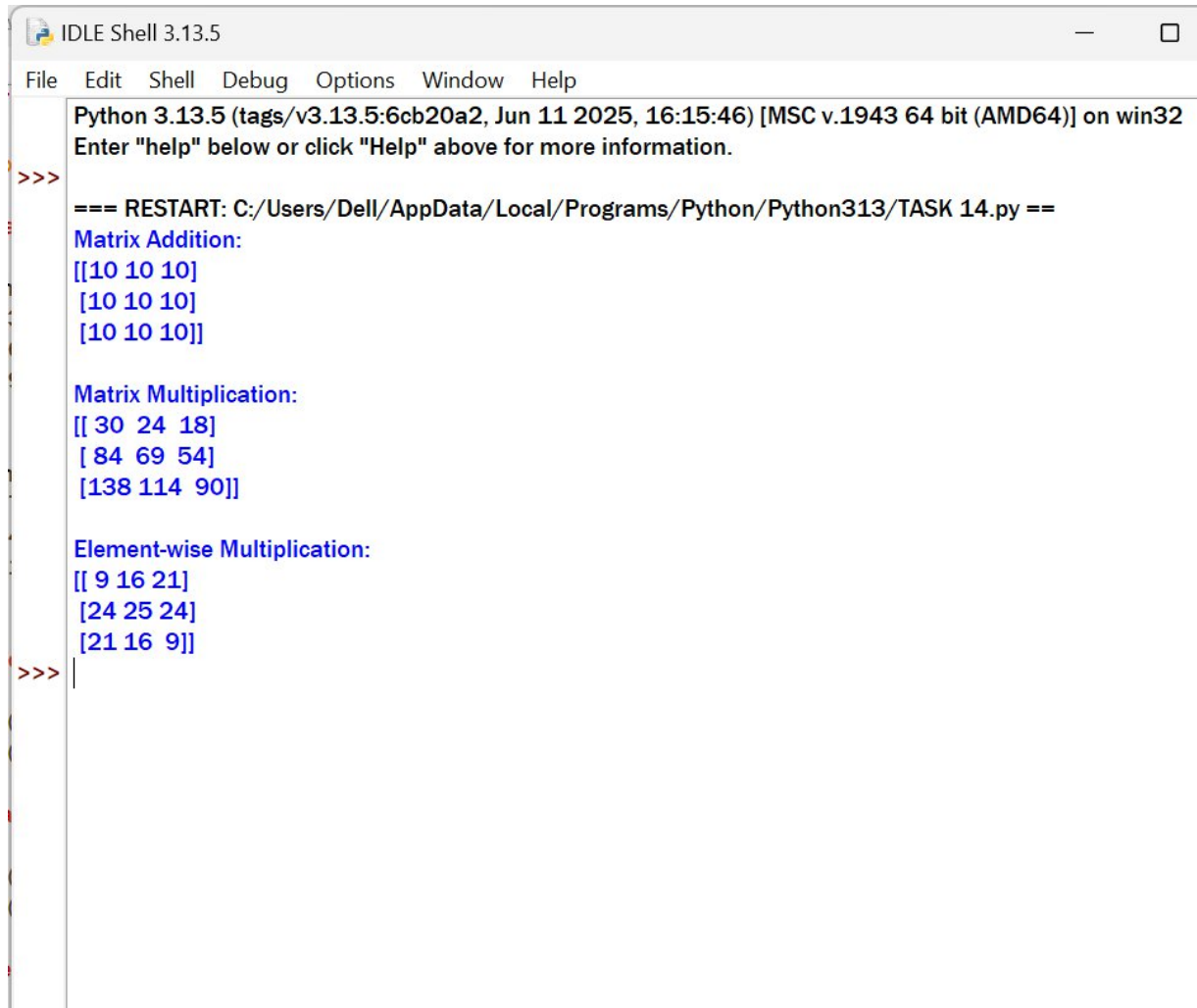
[ 2 12 30 56]

Dot Product:

100

&gt;&gt;&gt; |

Q7:



```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/TASK 14.py ===
Matrix Addition:
[[10 10 10]
 [10 10 10]
 [10 10 10]]

Matrix Multiplication:
[[ 30 24 18]
 [ 84 69 54]
 [138 114 90]]

Element-wise Multiplication:
[[ 9 16 21]
 [24 25 24]
 [21 16 9]]
>>> |
```

A large, light gray watermark with the word "SAMPLE" is oriented diagonally across the bottom half of the image.

### # Q7. (Matrices and Operations)

```
import numpy as np
```

#### # Creating matrices

```
A = np.array([
    [1,2,3],
    [4,5,6],
    [7,8,9]
])
```

```
B = np.array([
    [9,8,7],
    [6,5,4],
    [3,2,1]
])
```

#### # Addition

```
print("Matrix Addition:")
print(A + B)
```

#### # Matrix Multiplication

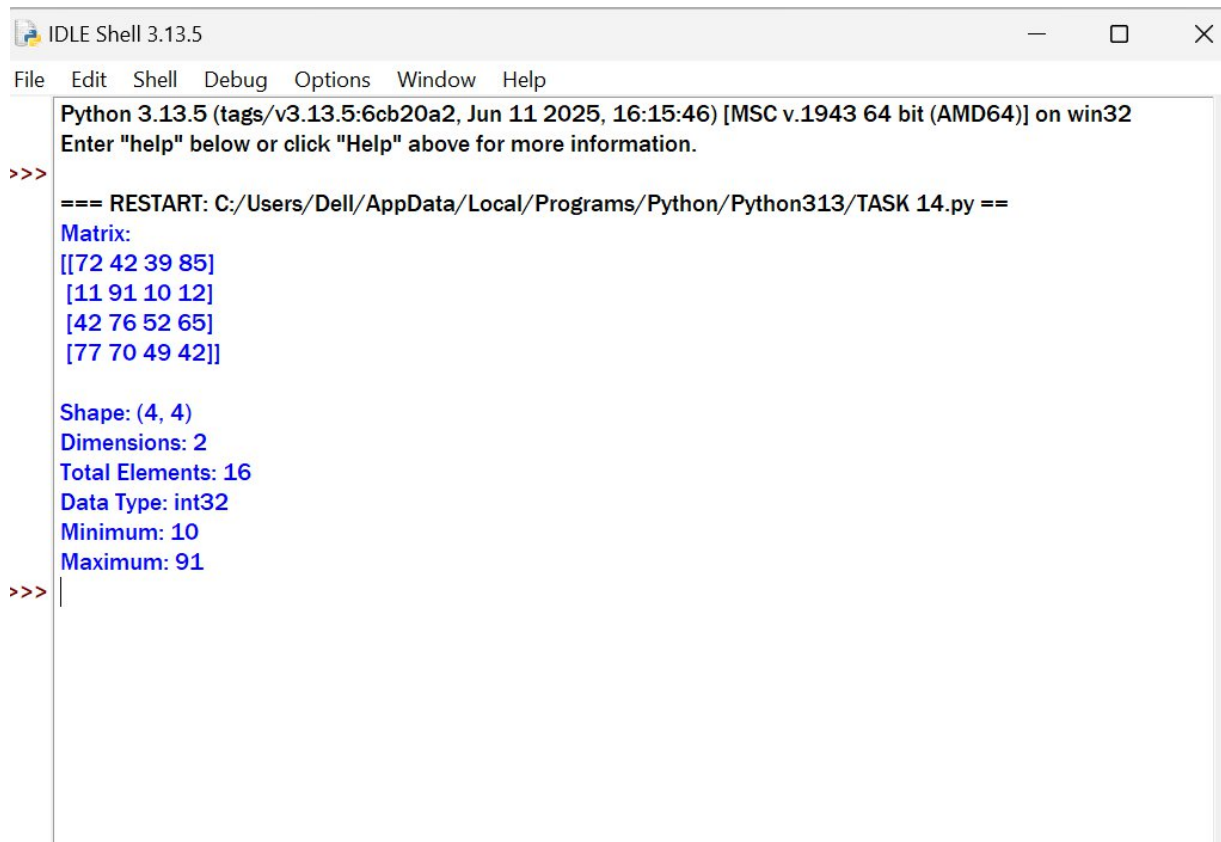
```
print("\nMatrix Multiplication:")
print(A @ B)
```

#### # Element-wise Multiplication

```
print("\nElement-wise Multiplication:")
print(A * B)
```

```
# End
```

Q8:



```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/TASK 14.py ===
Matrix:
[[72 42 39 85]
 [11 91 10 12]
 [42 76 52 65]
 [77 70 49 42]]

Shape: (4, 4)
Dimensions: 2
Total Elements: 16
Data Type: int32
Minimum: 10
Maximum: 91
>>> |
```

### # Q8. (Properties of Arrays)

```
import numpy as np
```

#### # Create random matrix

```
arr = np.random.randint(1, 101, (4,4))
```

```
print("Matrix:")
```

```
print(arr)
```

#### # Properties

```
print("\nShape:", arr.shape)
```

```
print("Dimensions:", arr.ndim)
```

```
print("Total Elements:", arr.size)
```

```
print("Data Type:", arr.dtype)
```

#### # Minimum and Maximum

```
print("Minimum:", arr.min())
```

```
print("Maximum:", arr.max())
```

```
# End
```

Q9:

TASK 14.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/TASK 14.py (3.13.5)

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**# Q9. (Random + Statistics)**

**import numpy as np**

**# Generate 20 random numbers**

**arr = np.random.randint(1, 51, 20)**

**print("Original Array:")**

**print(arr)**

**# Reshape**

**matrix = arr.reshape(4,5)**

**print("\n4x5 Matrix:")**

**print(matrix)**

**# Statistics**

**print("\nSum:", matrix.sum())**

**print("Mean:", matrix.mean())**

**print("Standard Deviation:", matrix.std())**

**# Maximum in each row**

**print("\nMaximum Value Row Wise:")**

**print(matrix.max(axis=1))**

**# End**

```
TASK 14.py IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on
Enter "help" below or click "Help" above for more information.

>>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/TASK 14.py ===
Original Array:
[24 44 47 12 11 24 10 27 2 17 48 46 15 20 37 32 10 48 34 32]

4x5 Matrix:
[[24 44 47 12 11]
 [24 10 27 2 17]
 [48 46 15 20 37]
 [32 10 48 34 32]]

Sum: 540
Mean: 27.0
Standard Deviation: 14.363147287415806

Maximum Value Row Wise:
[47 27 48 48]

>>>
print("\nSum:")
print("Mean:")
print("Standard Deviation:")

# Maximum in each row
print("\nMaximum in each row:")
print(matrix.max(axis=1))

# End
```

## Q10:

Task 13.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 13.py (3.13.5)

File Edit Format Run Options Window Help

Task 13.py

```
# Taking input
total = int(input("Enter number of random values to generate: "))

# Check input
if total <= 0:
    print("Enter value greater than 0")
else:

    # Generate random numbers
    numbers = np.random.randint(10, 101, total)

    print("\nGenerated Array:")
    print(numbers)

    # Statistics
    print("\nMean:", np.mean(numbers))
    print("Median:", np.median(numbers))
    print("Standard Deviation:", np.std(numbers))
    print("Minimum:", np.min(numbers))
    print("Maximum:", np.max(numbers))

    # Reshape if possible
    if total % 5 == 0:

        matrix = numbers.reshape(-1, 5)

        print("\n2D Matrix:")
        print(matrix)

        print("\nRow Wise Sum:")
        print(matrix.sum(axis=1))

        print(matrix.sum(axis=1))

    else:
        print("\nReshape not possible")

except ValueError:
    print("Invalid Input! Enter numbers only.")
```

```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 13.py ===
Enter number of random values to generate: 7

Generated Array:
[64 35 26 31 64 54 58]

Mean: 47.42857142857143
Median: 54.0
Standard Deviation: 15.060422521649658
Minimum: 26
Maximum: 64

Reshape not possible
>>>
```